

ACTL2102 Tutorial

Week 1: Stochastic Processes and Simulation

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Introduction

Hi! My name is **Nick**.

- ▶ 4th year Actuarial Studies and Computer Science student
- ▶ Took this course in T2 2024

How these tutorials work

- ▶ I'll go through the selected questions, and others on request.
- ▶ Before each question I'll briefly recap the relevant lecture concepts. This is **not** a replacement for the lectures.
- ▶ Week 1 is light on content; recaps will be quicker from next week, so watch the lectures first to get the most out of tutorials.

Plan

- ▶ Recap some content, then do the questions that use it
- ▶ Questions, easiest first: **1.10, 1.9, 1.3, 1.7, 1.8, 1.6**

What is a Stochastic Process?

A **stochastic process** is a collection of random variables $\{X(t), t \in T\}$.

- ▶ T is the **index set**, usually time
- ▶ $X(t)$ is the **state** at time t
- ▶ S is the **state space**, the set of values $X(t)$ can take

A **sample path** is one realisation of the whole process. One simulation run gives one sample path.

	Discrete time	Continuous time
Discrete state	Health status in week t	Alive or dead at time t
Continuous state	Share price at close of day t	Share price at time t

Modules 2–5 are almost entirely discrete-state: discrete time in Module 2, continuous time after that.

Increments

Independent increments

$X(t_0)$ and the increments

$$X(t_k) - X(t_{k-1}), \quad k = 1, \dots, n,$$

are independent for all

$$t_0 < \dots < t_n.$$

Future changes don't depend on the past.

Stationary increments

$$X(t_2 + \tau) - X(t_1 + \tau)$$

has the same distribution as $X(t_2) - X(t_1)$, for all $t_1 < t_2$ and $\tau > 0$.

Only the length of the interval matters.

Example. For a random walk $X_n = X_0 + \sum_{k=1}^n Z_k$ with X_0 fixed: independent Z_k give independent increments, and if they are also identically distributed, the increments are stationary too.

A **stationary process** is a stronger condition than stationary increments: the whole joint distribution is unchanged by shifting time.

The Markov Property

Write $X_k = X(t_k)$ for times $t_1 < t_2 < t_3$. For any process (multiplication rule, applied twice):

$$\Pr(X_1, X_2, X_3) = \Pr(X_1) \Pr(X_2 | X_1) \Pr(X_3 | X_2, X_1).$$

Markov: only the most recent value matters. In general, for any times $t_1 < \dots < t_{n+1}$,

$$\Pr(X_{n+1} | X_n, \dots, X_1) = \Pr(X_{n+1} | X_n),$$

so the joint distribution is built from $\Pr(X_1)$ and one-step probabilities alone.

Markov

Gambler's wealth

$X_{n+1} = X_n + Z_{n+1}$ with independent winnings. How you reached today's wealth is irrelevant: tomorrow's wealth depends only on today's wealth and Z_{n+1} .

Not Markov

Rain tomorrow depends on today *and* yesterday. This system is not Markovian, although it can be made Markovian by enlarging the state space.

Pseudo-Random Numbers

Every method below starts from $U(0,1)$ draws. A simple way to generate them:

Linear congruential generator. Pick a seed x_0 and positive integers a, c, m :

$$x_{n+1} = (ax_n + c) \bmod m, \quad u_n = x_n/m.$$

With $a = 11$, $c = 37$, $m = 100$ and $x_0 = 85$:

$$x_1 = (11 \cdot 85 + 37) \bmod 100 = 972 \bmod 100 = 72,$$

then $x_2 = 29$ and $x_3 = 56$, giving $u = 0.72, 0.29, 0.56$.

The x_n are integers; remember to divide by m .

Monte Carlo

To estimate $E[h(X)]$, simulate i.i.d. copies $X^{(1)}, \dots, X^{(r)}$ of X and average:

$$E[h(X)] \approx \frac{1}{r} \sum_{i=1}^r h(X^{(i)}).$$

By the law of large numbers, this converges to $E[h(X)]$ as $r \rightarrow \infty$.

Probabilities. Let $\mathbf{1}_A = 1$ if A happens, 0 otherwise. Then $E[\mathbf{1}_A] = \Pr(A)$, so

$\Pr(A) \approx$ proportion of simulations in which A happened.

Standard error

The estimate is itself random. Report its standard error $s/\sqrt{r}$, where s is the sample standard deviation of the values $h(X^{(i)})$.

Exercise 1.10: More than 3 heads in 10 flips

Exercise 1.10: [sim9, Solution] We flip a coin 10 times. Use Monte Carlo Simulation to derive the probability of getting more than 3 heads.

Integrals by Simulation

We want $I = \int_a^b h(x) dx$ where h has no closed-form antiderivative, e.g. $\int_0^1 e^{-x^2} dx$.

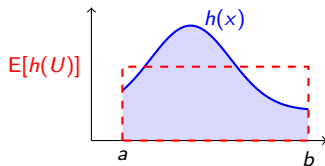
Monte Carlo estimates **expectations**, so we write I as one.

Take $U \sim U(a, b)$, e.g.

$U = a + (b - a)V$ with $V \sim U(0, 1)$.

Its density is constant, $\frac{1}{b-a}$, so

$E[h(U)]$ is the average height of h on $[a, b]$.



$$E[h(U)] = \int_a^b h(x) \frac{1}{b-a} dx = \frac{I}{b-a} \implies I = (b-a) E[h(U)].$$

Estimate

$I \approx (b-a) \cdot \frac{1}{r} \sum_{i=1}^r h(U_i)$, with independent $U_i \sim U(a, b)$.

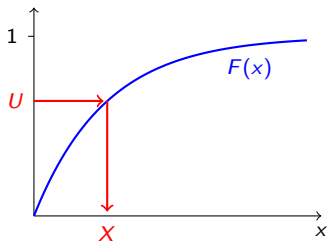
Example. Averaging $e^{-U_i^2}$ with $U_i \sim U(0, 1)$ gives

$$\int_0^1 e^{-x^2} dx \approx 0.7468.$$

Exercise 1.9: A normal integral by simulation

Exercise 1.9: [sim8, Solution] Suppose we have an Normal distribution with a mean of 1 and a standard deviation of 10. Find the integral $\int_3^6 \mathcal{N}(x; 1, 10)dx$ by simulation.

Inverse Transform



Method

1. Find the c.d.f. F .
2. Solve $u = F(x)$ for x .
3. Return $X = F^{-1}(U)$.

Why it works (for F continuous and strictly increasing)

Read across from U to F , then drop down to X .

$$\Pr(X \leq x) = \Pr(U \leq F(x)) = F(x).$$

Exponential: $u = 1 - e^{-\lambda x}$ gives $X = -\frac{1}{\lambda} \ln(1 - U)$.

Piecewise densities: invert each piece separately, and use the branch whose range of F contains U .

Exercise 1.3: Inverse transform for a two-piece density

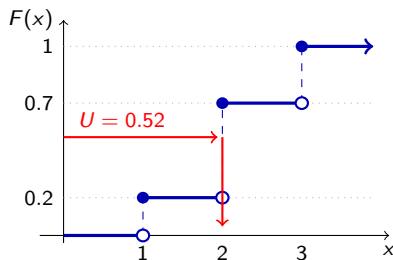
Exercise 1.3: [sim2, Solution] Let following function defines a valid probability density function:

$$h(x) = \begin{cases} k \cdot x^{-1/2}, & \text{if } 0 \leq x \leq 1; \\ k \cdot e^{-x}, & \text{if } 1 \leq x \leq \infty; \\ 0, & \text{otherwise.} \end{cases}$$

1. Describe how you would generate random values from this p.d.f. using the inverse transform method.
2. Apply the method to generate three random values from this distribution based on the pseudo-random values 0.6453, 0.2137 and 0.8672.

Discrete Inverse Transform

Take $X \in \{1, 2, 3\}$ with $\Pr(X = 1) = 0.2$, $\Pr(X = 2) = 0.5$, $\Pr(X = 3) = 0.3$. Its c.d.f. $F(x) = \Pr(X \leq x)$ is a step function:



If U falls in	return
$(0, 0.2]$	$X = 1$
$(0.2, 0.7]$	$X = 2$
$(0.7, 1]$	$X = 3$

Each interval's length is the probability of its value, the size of the jump there, so X gets the right distribution.

In general, for values $x_1 < x_2 < \dots$ with the convention $F(x_0) = 0$: $X = x_j$ when $F(x_{j-1}) < U \leq F(x_j)$, i.e.
 $X = \min\{x : F(x) \geq U\}$.

Geometric: for $\Pr(X = j) = p(1 - p)^{j-1}$, $j \geq 1$, this reduces to $X = \lceil \ln U / \ln(1 - p) \rceil$ (using U in place of $1 - U$).

Exercise 1.7: $\text{Bin}(5, 0.4)$ two ways

Exercise 1.7: [sim6, Solution]

1. Describe how you would simulate a $\text{Bin}(5, 0.4)$ distribution using the inverse transform method for discrete variates.
2. Apply the method to generate random values from this distribution based on the pseudo-random values 0.6453, 0.2137, 0.8672, 0.1426 and 0.4527.
3. Generate random values from this distribution using simulated values from a Bernoulli distribution based on the pseudo-random values 0.6453, 0.2137, 0.8672, 0.1426 and 0.4527.

Simulating a Path

A process $X_0, X_1, \dots, X_k$ moves one step at a time, and we want $E[V]$ for a value V computed from the path, such as the terminal value X_k or a payoff $\max(X_k - K, 0)$. Since X_k is built step by step, we simulate the **whole path**.

1. Simulate one full path. E.g. a gambler with $X_0 = 10$ wins 1 with probability 0.6 each step, else loses 1. Draw a fresh $U_n \sim U(0, 1)$ each step and move +1 if $U_n \leq 0.6$, otherwise -1:

Step n	1	2	3	4	5
u_n	0.31	0.78	0.05	0.66	0.42
X_n	11	10	11	10	11

With $k = 5$ and $V = X_5$, this path gives one value, $V = 11$.

2. Repeat. Simulate r independent full paths to get $V_1, \dots, V_r$, then estimate $E[V]$ by $\bar{V} = \frac{1}{r} \sum_i V_i$, with standard error $s/\sqrt{r}$ where s is the sample standard deviation of the V_i .

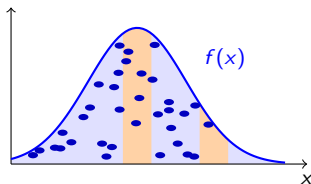
Exercise 1.8: Share price paths and a derivative payoff

Exercise 1.8: [sim7, Solution] A derivatives analyst models movements in share prices by assuming that on each day of trading the share price will either go up(U) by 1%, with probability p , or go down (D) by 1%, with probability $1 - p$. The value of p is assumed to be 0.52 during the next 13 weeks and 0.515 thereafter. The movements each day are independent of all previous movements.

1. Describe how you would create a simulated sequence of daily price movements for this model.
2. In one simulation the sequence of movements begins UUUUUUUUUUUUUUUUUUUUUUUUUUUUUUU...
If the initial share price is \$220, calculate the price at the end of the 25th trading day.
3. The profit arising from a particular derivative will be $\max(S - 220, 0)$ where S is the share price in one year's time (the 253rd trading day). Describe how you would use a simulation to estimate the mean and standard deviation of the profit arising from this product.

Acceptance–Rejection: The Idea

We want to simulate X with density f , but F^{-1} has no closed form, so inverse transform does not apply.



Key equivalence

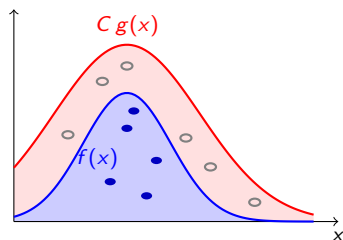
Simulating $X \sim f$ is the same as picking a point uniformly under the graph of f and taking its x -coordinate.

The strip over $[x, x + dx]$ has area $f(x) dx$ out of a total of 1, so taller parts of f are picked more often.

We cannot generate uniform points under f directly, but we can under Cg for a density g we already know how to sample.

Acceptance–Rejection: The Algorithm

Choose a density g we can sample and a constant C with $f(x) \leq C g(x)$ for all x .



• under f : accept ○ above f : reject

A uniform point under Cg :

take $Y \sim g$ as the x -position and $U \cdot Cg(Y)$, with $U \sim U(0, 1)$ independent of Y , as the height.

Its joint density is

$g(x) \cdot \frac{1}{Cg(x)} = \frac{1}{C}$, constant on the region under Cg .

Keep the point if it lies under f .
Kept points are uniform under f ,
so their x -values have density f .

Algorithm

Draw $Y \sim g$ and an independent $U \sim U(0, 1)$. If $U \leq f(Y)/(C g(Y))$, return $X = Y$; otherwise repeat.

Choosing C : a Normal from an Exponential

Use the smallest valid constant, $C = \sup_x f(x)/g(x)$, usually found by maximising f/g . The area under f is 1 and the area under Cg is C , so each attempt is accepted with probability $1/C$ and C attempts are needed on average. A g shaped like f keeps C small.

Example. Target $|Z|$, $Z \sim N(0, 1)$: $f(x) = \sqrt{2/\pi} e^{-x^2/2}$ for $x > 0$. Proposal $g(x) = e^{-x}$, i.e. $\text{Exp}(1)$. Completing the square,

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2e}{\pi}} e^{-(x-1)^2/2},$$

maximised at $x = 1$, so $C = \sqrt{2e/\pi} \approx 1.32$ and about 76% of draws are accepted.

1. Draw $U_1, U_2 \sim U(0, 1)$ and set $Y = -\ln U_1$.
2. Accept if $U_2 \leq e^{-(Y-1)^2/2}$; otherwise repeat.
3. Return $Z = \pm Y$ with an independent random sign, each with probability $\frac{1}{2}$.

Exercise 1.6: Beta–Binomial, parts 1–5

Exercise 1.6: [sim5, Solution] Consider a reinsurer who insures extreme event risks. By the nature of the insurance type, there are not many claims. The reinsurer models the number of claims in n years using a Binomial distribution(n, p). Hence, each year there is a probability p of getting a claim, for n consecutive years. Unfortunately, by the nature of the insurance product the estimate of the probability p is uncertain. Therefore, the probability p itself is a random variable and has a Beta(α, β) distribution. The uncertainty in the number of claims in n years, is called the Beta-Binomial distribution.

1. Explain in words what happens with the probability in the tails of the distribution when p is uncertain compared with the certain p , i.e., compare the tail of the Beta-Binomial distribution with the Binomial distribution.

2. Explain why the reinsurer is especially interested in the tail of a risk.

Hint: Think about the capital reserves insurers are required to hold.

Let m be the number of years of observation and k the number of years in the observation period with a claim, we have:

$$P \sim \text{Beta}(\alpha = m - k + 1, \beta = k + 1)$$

$$C \sim \text{Bin}(n, P),$$

where C is the random variable of the number of claims in n years. We have:

$$f_P(p) = b \cdot p^{\alpha-1} \cdot (1-p)^{\beta-1} \quad \text{with } 0 \leq p \leq 1$$
$$\text{where } b = \frac{1}{\int_0^1 p^{\alpha-1} \cdot (1-p)^{\beta-1}} = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \cdot \Gamma(\beta)}$$

First, consider the special case of $m = 2k$, i.e., $\alpha = \beta = k + 1 = m/2 + 1$.

3. Verify that the highest p.d.f. $f_P(p)$ would be achieved at $p = 1/2$.
4. Let $X \sim \text{Unif}(0, 1)$, with density $g(x) = 1$ if $0 \leq x \leq 1$ and zero otherwise. Prove that in the acceptance rejection method $c = b \cdot (1/4)^{n-1}$.

Hint: use the result in (3).

5. What would be the simulation procedure for simulating values of P ?

Now, consider the case where m and k are positive integers, where $m \geq k$.

6. Rewrite b such that $b = (m+1) \cdot \binom{m}{k}$.
7. Let $X \sim \text{Unif}(0, 1)$, with density $g(x) = 1$ if $0 \leq x \leq 1$ and zero otherwise. Show that $c = (m+1) \cdot \binom{m}{k} \cdot \left(1 - \frac{k}{m}\right)^{m-k} \cdot \left(\frac{k}{m}\right)^k$.
8. What would be the simulation procedure for simulating values of P ?
9. Comment on the efficiency of the simulation method for large values of m .

Common Mistakes

- ▶ **Forgetting to divide by m** in an LCG.
- ▶ **Dropping the width $(b - a)$** when estimating an integral.
- ▶ **Mixing up U and $1 - U$** . Both are uniform, so either works, but say which you use.
- ▶ **Getting the discrete inequality backwards**. It is $F(x_{j-1}) < U \leq F(x_j)$.
- ▶ **Treating one simulated path as the answer**. You need many independent paths.
- ▶ **Choosing C that doesn't cover f everywhere**. $Cg \geq f$ must hold for all x .

Thanks for coming!